Customer Feedback Analysis using Logistic Regression

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Contents

```r
if (!"pacman" %in% installed.packages()[, "Package"]) {
  install.packages("pacman", dependencies = TRUE)
  library("pacman", character.only = TRUE)
}

pacman::p_load("here")

knitr::opts_knit$set(root.dir = here::here())
```

1 Load the Dataset

The synthetic customer feedback dataset contains 10,888 observations and 43 variables. The variables that form the predictors are the following 4:

Table 1: List of predictors

Column Number	Predictor
19	customerfeedback_foodquality
20	customerfeedback_servicequality
21	customerfeedback_pricetovalue
22	customerfeedback_ambiance

And the variable that is the outcome is column 11, i.e., `customer_status`

```r
pacman::p_load("readr")

customer_feedback_data <-
  read_csv("data/siwaka_dishes_view_customerfeedback_data.csv", col_types = cols(
    customer_status = col_factor(levels = c("1", "0")),
    customerorder_orderDate = col_datetime(format = "%Y-%m-%d %H:%M:%S"),
    customerorder_requiredDate = col_datetime(format = "%Y-%m-%d %H:%M:%S"),
    customerorder_dispatchDate = col_datetime(format = "%Y-%m-%d %H:%M:%S")
    )
```

```r
  )

head(customer_feedback_data)
```

```
## # A tibble: 6 x 43
##   customer_customerNumber customer_customerName       customer_contactFirstName
##                     <dbl> <chr>                       <chr>
## 1                     111 Achieng Ochieng             Achieng
## 2                     491 Michel Bizimana             Michel
## 3                     491 Michel Bizimana             Michel
## 4                     491 Michel Bizimana             Michel
## 5                     711 [Business] Ole Sereni Lodge Mwangi
## 6                     711 [Business] Ole Sereni Lodge Mwangi
## # i 40 more variables: customer_contactLastName <chr>, customer_phone <chr>,
## #   customer_addressLine1 <chr>, customer_addressLine2 <chr>,
## #   customer_postalCode <dbl>, customer_county <chr>, customer_subCounty <chr>,
## #   customer_status <fct>, customerorder_orderNumber <dbl>,
## #   customerorder_orderDate <dttm>, customerorder_requiredDate <dttm>,
## #   customerorder_dispatchDate <dttm>, customerorder_orderStatusID <dbl>,
## #   customerorder_customerNumber <dbl>, ...
```

2 Initial EDA

View the Dimensions

The number of observations and the number of variables.

```r
dim(customer_feedback_data)
```

```
## [1] 10888    43
```

View the Data Types

```r
sapply(customer_feedback_data, class)
```

```
## $customer_customerNumber
## [1] "numeric"
##
## $customer_customerName
## [1] "character"
##
## $customer_contactFirstName
## [1] "character"
##
## $customer_contactLastName
## [1] "character"
##
## $customer_phone
## [1] "character"
##
```

```
## $customer_addressLine1
## [1] "character"
##
## $customer_addressLine2
## [1] "character"
##
## $customer_postalCode
## [1] "numeric"
##
## $customer_county
## [1] "character"
##
## $customer_subCounty
## [1] "character"
##
## $customer_status
## [1] "factor"
##
## $customerorder_orderNumber
## [1] "numeric"
##
## $customerorder_orderDate
## [1] "POSIXct" "POSIXt"
##
## $customerorder_requiredDate
## [1] "POSIXct" "POSIXt"
##
## $customerorder_dispatchDate
## [1] "POSIXct" "POSIXt"
##
## $customerorder_orderStatusID
## [1] "numeric"
##
## $customerorder_customerNumber
## [1] "numeric"
##
## $customerfeedback_customerfeedbackID
## [1] "numeric"
##
## $customerfeedback_foodquality
## [1] "numeric"
##
## $customerfeedback_servicequality
## [1] "numeric"
##
## $customerfeedback_pricetovalue
## [1] "numeric"
##
## $customerfeedback_ambiance
## [1] "numeric"
##
## $customerfeedback_comment
## [1] "character"
##
```

```
## $orderdetail_productCode
## [1] "character"
##
## $orderdetail_quantityOrdered
## [1] "numeric"
##
## $orderdetail_priceEach
## [1] "numeric"
##
## $product_productName
## [1] "character"
##
## $product_productDescription
## [1] "character"
##
## $product_quantityInStock
## [1] "numeric"
##
## $product_costOfProduction
## [1] "numeric"
##
## $product_sellingPrice
## [1] "numeric"
##
## $productcategory_productCategoryID
## [1] "numeric"
##
## $productcategory_categoryName
## [1] "character"
##
## $productcategory_categoryDescription
## [1] "character"
##
## $orderstatus_orderStatusID
## [1] "numeric"
##
## $orderstatus_status
## [1] "character"
##
## $branch_branchCode
## [1] "numeric"
##
## $branch_phone
## [1] "character"
##
## $branch_addressLine1
## [1] "character"
##
## $branch_addressLine2
## [1] "character"
##
## $branch_postalCode
## [1] "character"
##
```

```
## $branch_county
## [1] "character"
##
## $branch_subCounty
## [1] "character"
```

```r
str(customer_feedback_data)
```

```
## spc_tbl_ [10,888 x 43] (S3: spec_tbl_df/tbl_df/tbl/data.frame)
##  $ customer_customerNumber           : num [1:10888] 111 491 491 491 711 711 121 266 266 266 ...
##  $ customer_customerName             : chr [1:10888] "Achieng Ochieng" "Michel Bizimana" "Michel Bi
##  $ customer_contactFirstName         : chr [1:10888] "Achieng" "Michel" "Michel" "Michel" ...
##  $ customer_contactLastName          : chr [1:10888] "Ochieng" "Bizimana" "Bizimana" "Bizimana" ...
##  $ customer_phone                    : chr [1:10888] "0762556460" "0733475327" "0733475327" "073347
##  $ customer_addressLine1             : chr [1:10888] "993 Uhuru Hwy" "849 Uhuru Hwy" "849 Uhuru Hwy
##  $ customer_addressLine2             : chr [1:10888] "Apt 6" "Apt 16" "Apt 16" "Apt 16" ...
##  $ customer_postalCode               : num [1:10888] 19399 84540 84540 84540 19764 ...
##  $ customer_county                   : chr [1:10888] "Mombasa" "Kisumu" "Kisumu" "Kisumu" ...
##  $ customer_subCounty                : chr [1:10888] "Moiben" "Kisauni" "Kisauni" "Kisauni" ...
##  $ customer_status                   : Factor w/ 2 levels "1","0": 2 1 1 1 1 1 2 1 1 1 ...
##  $ customerorder_orderNumber         : num [1:10888] 4 29 29 29 40 40 53 62 62 62 ...
##  $ customerorder_orderDate           : POSIXct[1:10888], format: "2023-06-22 04:47:02" "2025-01-10 
##  $ customerorder_requiredDate        : POSIXct[1:10888], format: "2023-06-22 11:09:36" "2025-01-11 
##  $ customerorder_dispatchDate        : POSIXct[1:10888], format: "2023-06-22 10:17:33" "2025-01-16 
##  $ customerorder_orderStatusID       : num [1:10888] 2 3 3 3 1 1 2 4 4 4 ...
##  $ customerorder_customerNumber      : num [1:10888] 111 491 491 491 711 711 121 266 266 266 ...
##  $ customerfeedback_customerfeedbackID: num [1:10888] 808 3546 3546 3546 5049 ...
##  $ customerfeedback_foodquality      : num [1:10888] 2 4 4 4 4 4 3 4 4 4 ...
##  $ customerfeedback_servicequality   : num [1:10888] 3 5 5 5 4 4 4 4 4 4 ...
##  $ customerfeedback_pricetovalue     : num [1:10888] 2 4 4 4 3 3 4 4 4 4 ...
##  $ customerfeedback_ambiance         : num [1:10888] 3 4 4 4 3 3 4 4 4 4 ...
##  $ customerfeedback_comment          : chr [1:10888] "Limited vegetarian options." "Staff were very
##  $ orderdetail_productCode           : chr [1:10888] "P053" "P090" "P053" "P035" ...
##  $ orderdetail_quantityOrdered       : num [1:10888] 1 4 1 6 7 4 8 5 5 1 ...
##  $ orderdetail_priceEach             : num [1:10888] 70 50 70 70 70 150 50 200 70 60 ...
##  $ product_productName               : chr [1:10888] "Matoke na Sukuma" "Akara" "Matoke na Sukuma" 
##  $ product_productDescription        : chr [1:10888] "Plantains cooked with collard greens." "Deep-
##  $ product_quantityInStock           : num [1:10888] 70 100 70 70 70 60 100 70 70 90 ...
##  $ product_costOfProduction          : num [1:10888] 30 20 30 30 30 70 20 100 30 30 ...
##  $ product_sellingPrice              : num [1:10888] 70 50 70 70 70 150 50 200 70 60 ...
##  $ productcategory_productCategoryID : num [1:10888] 6 10 6 6 6 3 1 3 6 5 ...
##  $ productcategory_categoryName      : chr [1:10888] "Soup/Stew Dishes" "Combination Plates" "Soup/
##  $ productcategory_categoryDescription: chr [1:10888] "Soups or stews made with a variety of ingredi
##  $ orderstatus_orderStatusID         : num [1:10888] 2 3 3 3 1 1 2 4 4 4 ...
##  $ orderstatus_status                : chr [1:10888] "Processing" "In Transit" "In Transit" "In Tra
##  $ branch_branchCode                 : num [1:10888] 1 1 1 1 1 1 1 1 1 1 ...
##  $ branch_phone                      : chr [1:10888] "0700000001" "0700000001" "0700000001" "070000
##  $ branch_addressLine1               : chr [1:10888] "Westlands Commercial Centre" "Westlands Comme
##  $ branch_addressLine2               : chr [1:10888] "Ring Road, Westlands" "Ring Road, Westlands" 
##  $ branch_postalCode                 : chr [1:10888] "00100" "00100" "00100" "00100" ...
##  $ branch_county                     : chr [1:10888] "Nairobi" "Nairobi" "Nairobi" "Nairobi" ...
##  $ branch_subCounty                  : chr [1:10888] "Westlands" "Westlands" "Westlands" "Westlands
##  - attr(*, "spec")=
##   .. cols(
```

```
##   ..   customer_customerNumber = col_double(),
##   ..   customer_customerName = col_character(),
##   ..   customer_contactFirstName = col_character(),
##   ..   customer_contactLastName = col_character(),
##   ..   customer_phone = col_character(),
##   ..   customer_addressLine1 = col_character(),
##   ..   customer_addressLine2 = col_character(),
##   ..   customer_postalCode = col_double(),
##   ..   customer_county = col_character(),
##   ..   customer_subCounty = col_character(),
##   ..   customer_status = col_factor(levels = c("1", "0"), ordered = FALSE, include_na = FALSE),
##   ..   customerorder_orderNumber = col_double(),
##   ..   customerorder_orderDate = col_datetime(format = "%Y-%m-%d %H:%M:%S"),
##   ..   customerorder_requiredDate = col_datetime(format = "%Y-%m-%d %H:%M:%S"),
##   ..   customerorder_dispatchDate = col_datetime(format = "%Y-%m-%d %H:%M:%S"),
##   ..   customerorder_orderStatusID = col_double(),
##   ..   customerorder_customerNumber = col_double(),
##   ..   customerfeedback_customerfeedbackID = col_double(),
##   ..   customerfeedback_foodquality = col_double(),
##   ..   customerfeedback_servicequality = col_double(),
##   ..   customerfeedback_pricetovalue = col_double(),
##   ..   customerfeedback_ambiance = col_double(),
##   ..   customerfeedback_comment = col_character(),
##   ..   orderdetail_productCode = col_character(),
##   ..   orderdetail_quantityOrdered = col_double(),
##   ..   orderdetail_priceEach = col_double(),
##   ..   product_productName = col_character(),
##   ..   product_productDescription = col_character(),
##   ..   product_quantityInStock = col_double(),
##   ..   product_costOfProduction = col_double(),
##   ..   product_sellingPrice = col_double(),
##   ..   productcategory_productCategoryID = col_double(),
##   ..   productcategory_categoryName = col_character(),
##   ..   productcategory_categoryDescription = col_character(),
##   ..   orderstatus_orderStatusID = col_double(),
##   ..   orderstatus_status = col_character(),
##   ..   branch_branchCode = col_double(),
##   ..   branch_phone = col_character(),
##   ..   branch_addressLine1 = col_character(),
##   ..   branch_addressLine2 = col_character(),
##   ..   branch_postalCode = col_character(),
##   ..   branch_county = col_character(),
##   ..   branch_subCounty = col_character()
##   .. )
##  - attr(*, "problems")=<externalptr>
```

Descriptive Statistics

2.1 Measures of Frequency

49.92% of the feedback was from customers who are active and 50.08% of the feedback was from customers who are dormant. The distribution of observations amongst the two classes was therefore balanced.

```r
customer_feedback_data_freq <- customer_feedback_data$customer_status
cbind(frequency = table(customer_feedback_data_freq),
      percentage = prop.table(table(customer_feedback_data_freq)) * 100)
```

```
##   frequency percentage
## 1      5435   49.91734
## 0      5453   50.08266
```

2.2 Measures of Central Tendency

The median and the mean of each numeric variable:

```r
summary(customer_feedback_data)
```

```
##  customer_customerNumber customer_customerName customer_contactFirstName
##  Min.   :  1.0           Length:10888          Length:10888
##  1st Qu.:190.0           Class :character      Class :character
##  Median :377.0           Mode  :character      Mode  :character
##  Mean   :381.7
##  3rd Qu.:574.0
##  Max.   :770.0
##
##  customer_contactLastName customer_phone     customer_addressLine1
##  Length:10888             Length:10888       Length:10888
##  Class :character         Class :character   Class :character
##  Mode  :character         Mode  :character   Mode  :character
##
##
##
##
##  customer_addressLine2 customer_postalCode customer_county
##  Length:10888          Min.   :  377       Length:10888
##  Class :character      1st Qu.:22731       Class :character
##  Mode  :character      Median :44424       Mode  :character
##                        Mean   :45002
##                        3rd Qu.:67127
##                        Max.   :90412
##
##  customer_subCounty customer_status customerorder_orderNumber
##  Length:10888       1:5435          Min.   :   1
##  Class :character   0:5453          1st Qu.:1359
##  Mode  :character                   Median :2746
##                                     Mean   :2742
##                                     3rd Qu.:4114
##                                     Max.   :5500
##
##  customerorder_orderDate       customerorder_requiredDate
##  Min.   :2021-01-01 03:34:53   Min.   :2021-01-03 05:55:28
##  1st Qu.:2022-02-05 15:17:54   1st Qu.:2022-02-08 09:44:23
##  Median :2023-03-10 21:21:58   Median :2023-03-13 20:42:37
##  Mean   :2023-03-13 18:41:52   Mean   :2023-03-16 05:56:07
##  3rd Qu.:2024-04-19 06:02:59   3rd Qu.:2024-04-20 17:11:06
```

```
##  Max.   :2025-05-30 20:02:21   Max.   :2025-06-03 18:42:08
##
##  customerorder_dispatchDate    customerorder_orderStatusID
##  Min.   :2021-01-01 23:13:52   Min.   :1.000
##  1st Qu.:2022-02-05 03:16:34   1st Qu.:2.000
##  Median :2023-03-11 13:32:27   Median :3.000
##  Mean   :2023-03-12 21:22:50   Mean   :2.993
##  3rd Qu.:2024-04-18 02:33:47   3rd Qu.:4.000
##  Max.   :2025-06-04 23:16:00   Max.   :5.000
##  NA's   :2066
##  customerorder_customerNumber customerfeedback_customerfeedbackID
##  Min.   :  1.0                Min.   :   1
##  1st Qu.:190.0                1st Qu.:1372
##  Median :377.0                Median :2739
##  Mean   :381.7                Mean   :2744
##  3rd Qu.:574.0                3rd Qu.:4110
##  Max.   :770.0                Max.   :5500
##
##  customerfeedback_foodquality customerfeedback_servicequality
##  Min.   :1.00                 Min.   :1.000
##  1st Qu.:2.00                 1st Qu.:2.000
##  Median :3.00                 Median :3.000
##  Mean   :2.93                 Mean   :2.933
##  3rd Qu.:4.00                 3rd Qu.:4.000
##  Max.   :5.00                 Max.   :5.000
##
##  customerfeedback_pricetovalue customerfeedback_ambiance
##  Min.   :1.000                 Min.   :1.000
##  1st Qu.:2.000                 1st Qu.:2.000
##  Median :3.000                 Median :3.000
##  Mean   :2.939                 Mean   :2.929
##  3rd Qu.:4.000                 3rd Qu.:4.000
##  Max.   :5.000                 Max.   :5.000
##
##  customerfeedback_comment orderdetail_productCode orderdetail_quantityOrdered
##  Length:10888             Length:10888            Min.   :1.000
##  Class :character         Class :character        1st Qu.:2.000
##  Mode  :character         Mode  :character        Median :4.000
##                                                   Mean   :4.481
##                                                   3rd Qu.:6.000
##                                                   Max.   :8.000
##
##  orderdetail_priceEach product_productName product_productDescription
##  Min.   : 20.0         Length:10888        Length:10888
##  1st Qu.: 60.0         Class :character    Class :character
##  Median : 70.0         Mode  :character    Mode  :character
##  Mean   :113.3
##  3rd Qu.:200.0
##  Max.   :400.0
##
##  product_quantityInStock product_costOfProduction product_sellingPrice
##  Min.   : 50.00          Min.   :  5.00           Min.   : 20.0
##  1st Qu.: 60.00          1st Qu.: 30.00           1st Qu.: 60.0
##  Median : 70.00          Median : 30.00           Median : 70.0
```

```
##  Mean   : 76.51         Mean   : 54.65          Mean   :113.3
##  3rd Qu.: 90.00         3rd Qu.:100.00          3rd Qu.:200.0
##  Max.   :100.00         Max.   :200.00          Max.   :400.0
##
##  productcategory_productCategoryID productcategory_categoryName
##  Min.   : 1.000                    Length:10888
##  1st Qu.: 3.000                    Class :character
##  Median : 6.000                    Mode  :character
##  Mean   : 6.203
##  3rd Qu.:10.000
##  Max.   :11.000
##
##  productcategory_categoryDescription orderstatus_orderStatusID
##  Length:10888                        Min.   :1.000
##  Class :character                    1st Qu.:2.000
##  Mode  :character                    Median :3.000
##                                      Mean   :2.993
##                                      3rd Qu.:4.000
##                                      Max.   :5.000
##
##  orderstatus_status branch_branchCode branch_phone       branch_addressLine1
##  Length:10888       Min.   : 1.00     Length:10888       Length:10888
##  Class :character   1st Qu.: 6.00     Class :character   Class :character
##  Mode  :character   Median :11.00     Mode  :character   Mode  :character
##                     Mean   :10.56
##                     3rd Qu.:15.00
##                     Max.   :20.00
##
##  branch_addressLine2 branch_postalCode  branch_county      branch_subCounty
##  Length:10888        Length:10888       Length:10888       Length:10888
##  Class :character    Class :character   Class :character   Class :character
##  Mode  :character    Mode  :character   Mode  :character   Mode  :character
##
##
##
##
```

2.3 Measures of Distribution

Measuring the variability in the dataset is important because the amount of variability determines **how well you can generalize** results from the sample to a new observation in the population.

Low variability is ideal because it means that you can better predict information about the population based on the sample data. High variability means that the values are less consistent, thus making it harder to make predictions.

2.3.1 Variance

```r
sapply(customer_feedback_data[,c(19,20,21,22)], var)
```

```
##    customerfeedback_foodquality customerfeedback_servicequality
```

```
##                        1.723978                          1.823318
##    customerfeedback_pricetovalue       customerfeedback_ambiance
##                        1.778164                          1.807357
```

2.3.2 Standard Deviation

```r
sapply(customer_feedback_data[, c(19,20,21,22)], sd)
```

```
##     customerfeedback_foodquality customerfeedback_servicequality
##                        1.313004                          1.350303
##    customerfeedback_pricetovalue       customerfeedback_ambiance
##                        1.333478                          1.344380
```

2.3.3 Kurtosis

The Kurtosis informs us of how often outliers occur in the results. There are different formulas for calculating kurtosis. Specifying "type = 2" allows us to use the 2nd formula which is the same kurtosis formula used in other statistical software like SPSS and SAS.

In "type = 2" (used in SPSS and SAS):

1. Kurtosis < 3 implies a low number of outliers
2. Kurtosis = 3 implies a medium number of outliers
3. Kurtosis > 3 implies a high number of outliers

```r
pacman::p_load("e1071")
sapply(customer_feedback_data[,c(19,20,21,22)],  kurtosis, type = 2)
```

```
##     customerfeedback_foodquality customerfeedback_servicequality
##                       -1.124285                         -1.165533
##    customerfeedback_pricetovalue       customerfeedback_ambiance
##                       -1.144930                         -1.166783
```

2.3.4 Skewness

The skewness is used to identify the asymmetry of the distribution of results. Similar to kurtosis, there are several ways of computing the skewness.

Using "type = 2" (common in other statistical software like SPSS and SAS) can be interpreted as:

1. Skewness between -0.4 and 0.4 (inclusive) implies that there is no skew in the distribution of results; the distribution of results is symmetrical; it is a normal distribution; a Gaussian distribution.
2. Skewness above 0.4 implies a positive skew; a right-skewed distribution.
3. Skewness below -0.4 implies a negative skew; a left-skewed distribution.

```r
sapply(customer_feedback_data[,c(19,20,21,22)], skewness, type = 2)
```

```
##    customerfeedback_foodquality customerfeedback_servicequality
##                      0.06355783                      0.06449936
##   customerfeedback_pricetovalue       customerfeedback_ambiance
##                      0.05552955                      0.06522520
```

2.4 Measures of Relationship

2.4.1 Covariance

Covariance is a statistical measure that indicates the direction of the linear relationship between two variables. It assesses whether increases in one variable correspond to increases or decreases in another.

- **Positive Covariance:** When one variable increases, the other tends to increase as well.
- **Negative Covariance:** When one variable increases, the other tends to decrease.
- **Zero Covariance:** No linear relationship exists between the variables.

While covariance indicates the direction of a relationship, it does not convey the strength or consistency of the relationship. The correlation coefficient is used to indicate the strength of the relationship.

```r
cov(customer_feedback_data[,c(19,20,21,22)], method = "spearman")
```

```
##                                 customerfeedback_foodquality
## customerfeedback_foodquality                         9451173
## customerfeedback_servicequality                      8564585
## customerfeedback_pricetovalue                        8546703
## customerfeedback_ambiance                            8581672
##                                 customerfeedback_servicequality
## customerfeedback_foodquality                            8564585
## customerfeedback_servicequality                         9465764
## customerfeedback_pricetovalue                           8055767
## customerfeedback_ambiance                               8080093
##                                 customerfeedback_pricetovalue
## customerfeedback_foodquality                          8546703
## customerfeedback_servicequality                       8055767
## customerfeedback_pricetovalue                         9461237
## customerfeedback_ambiance                             8067989
##                                 customerfeedback_ambiance
## customerfeedback_foodquality                      8581672
## customerfeedback_servicequality                   8080093
## customerfeedback_pricetovalue                     8067989
## customerfeedback_ambiance                         9466306
```

2.4.2 Correlation

A strong correlation between variables enables us to better predict the value of the dependent variable using the value of the independent variable. However, a weak correlation between two variables does not help us to predict the value of the dependent variable from the value of the independent variable. This is useful only if there is a linear association between the variables.

We can measure the statistical significance of the correlation using Spearman's rank correlation *rho*. This shows us if the variables are significantly monotonically related. A monotonic relationship between two variables implies that as one variable increases, the other variable either consistently increases or consistently decreases. The key characteristic is the preservation of the direction of change, though the rate of change may vary.

To view the correlation of all variables at the same time

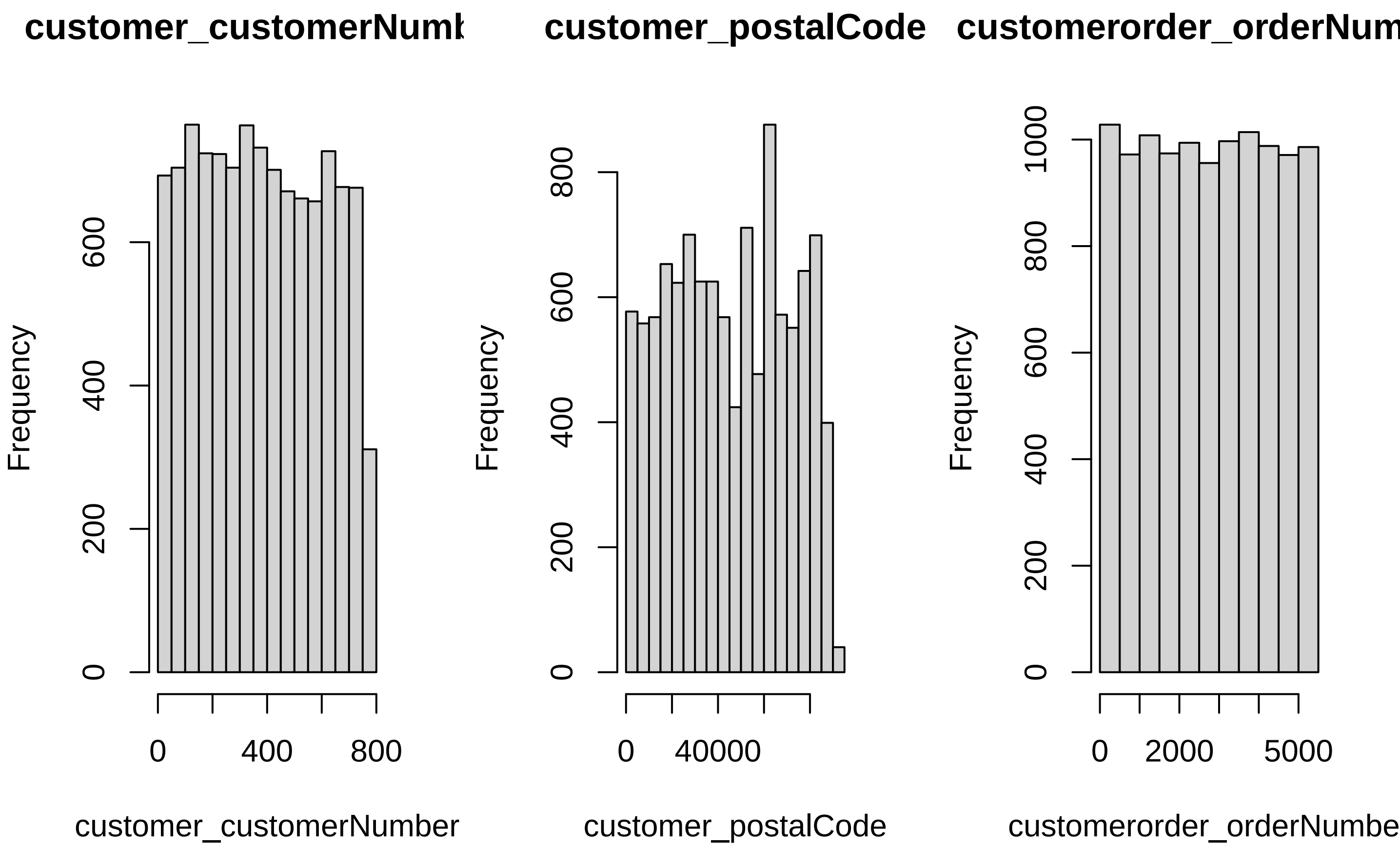
```
cor(customer_feedback_data[,c(19,20,21,22)], method = "spearman")
```

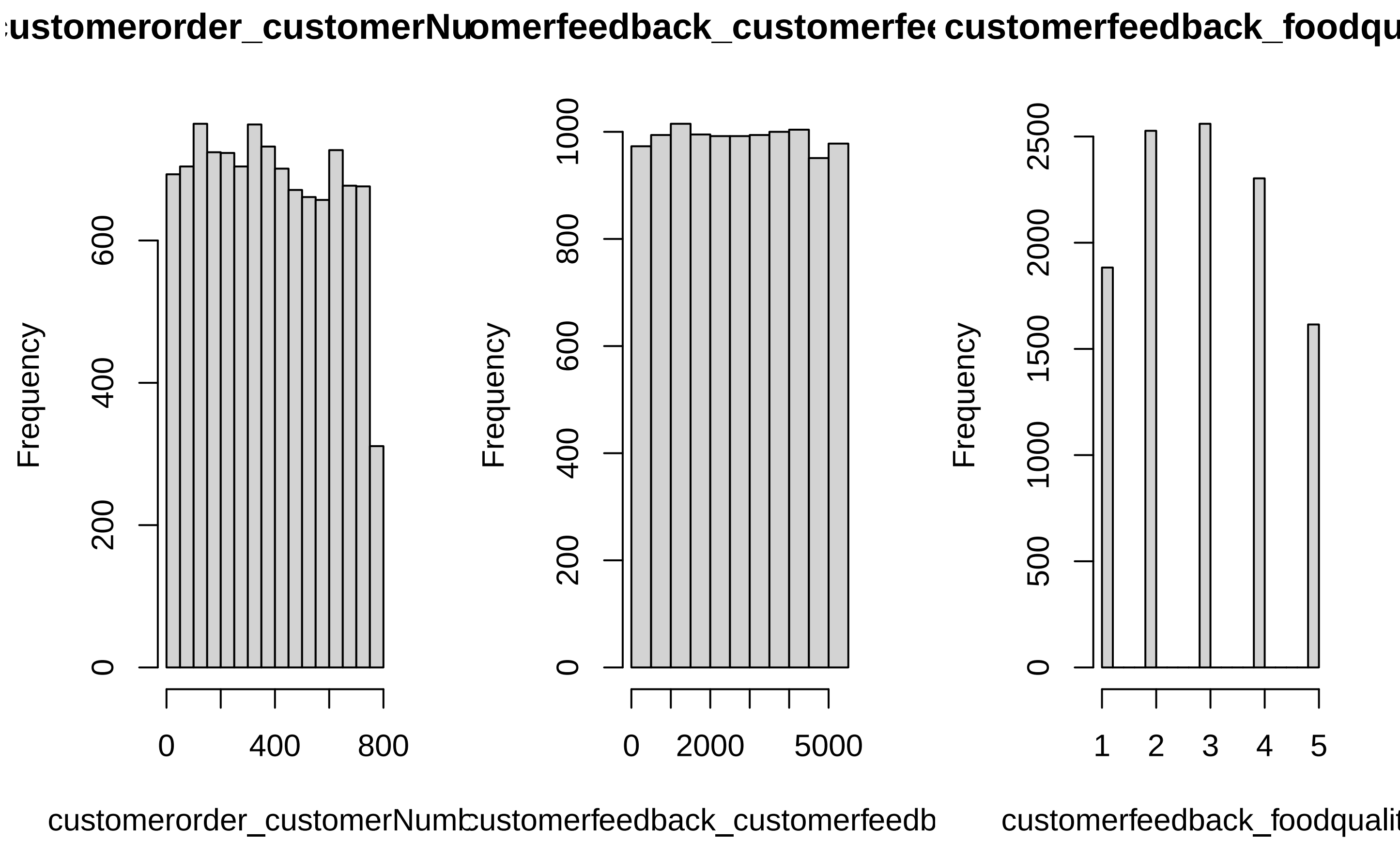
```
##                                customerfeedback_foodquality
## customerfeedback_foodquality                      1.0000000
## customerfeedback_servicequality                   0.9054941
## customerfeedback_pricetovalue                     0.9038196
## customerfeedback_ambiance                         0.9072746
##                                customerfeedback_servicequality
## customerfeedback_foodquality                         0.9054941
## customerfeedback_servicequality                      1.0000000
## customerfeedback_pricetovalue                        0.8512460
## customerfeedback_ambiance                            0.8535879
##                                customerfeedback_pricetovalue
## customerfeedback_foodquality                       0.9038196
## customerfeedback_servicequality                    0.8512460
## customerfeedback_pricetovalue                      1.0000000
## customerfeedback_ambiance                          0.8525131
##                                customerfeedback_ambiance
## customerfeedback_foodquality                   0.9072746
## customerfeedback_servicequality                0.8535879
## customerfeedback_pricetovalue                  0.8525131
## customerfeedback_ambiance                      1.0000000
```

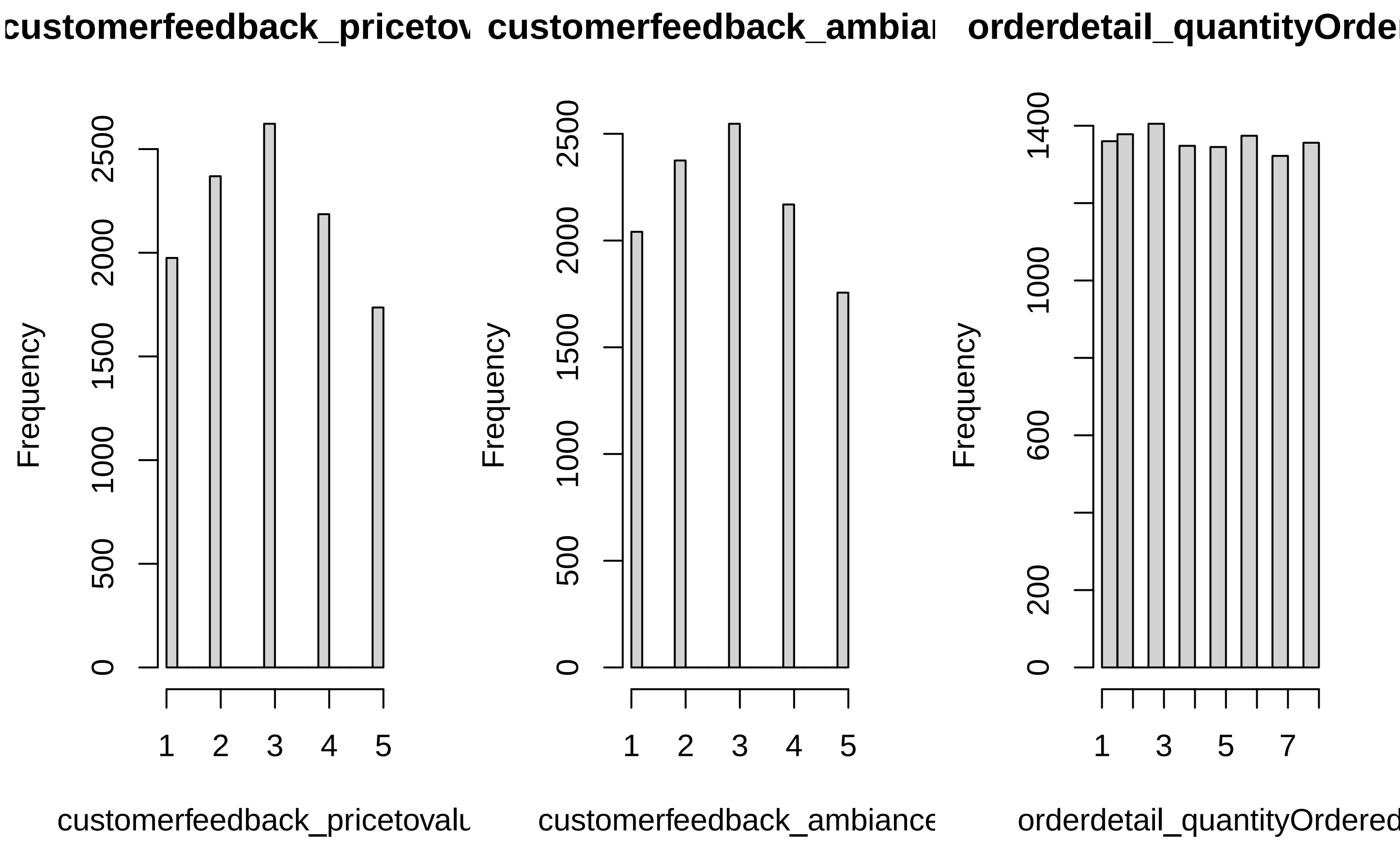
2.5 Basic Visualizations

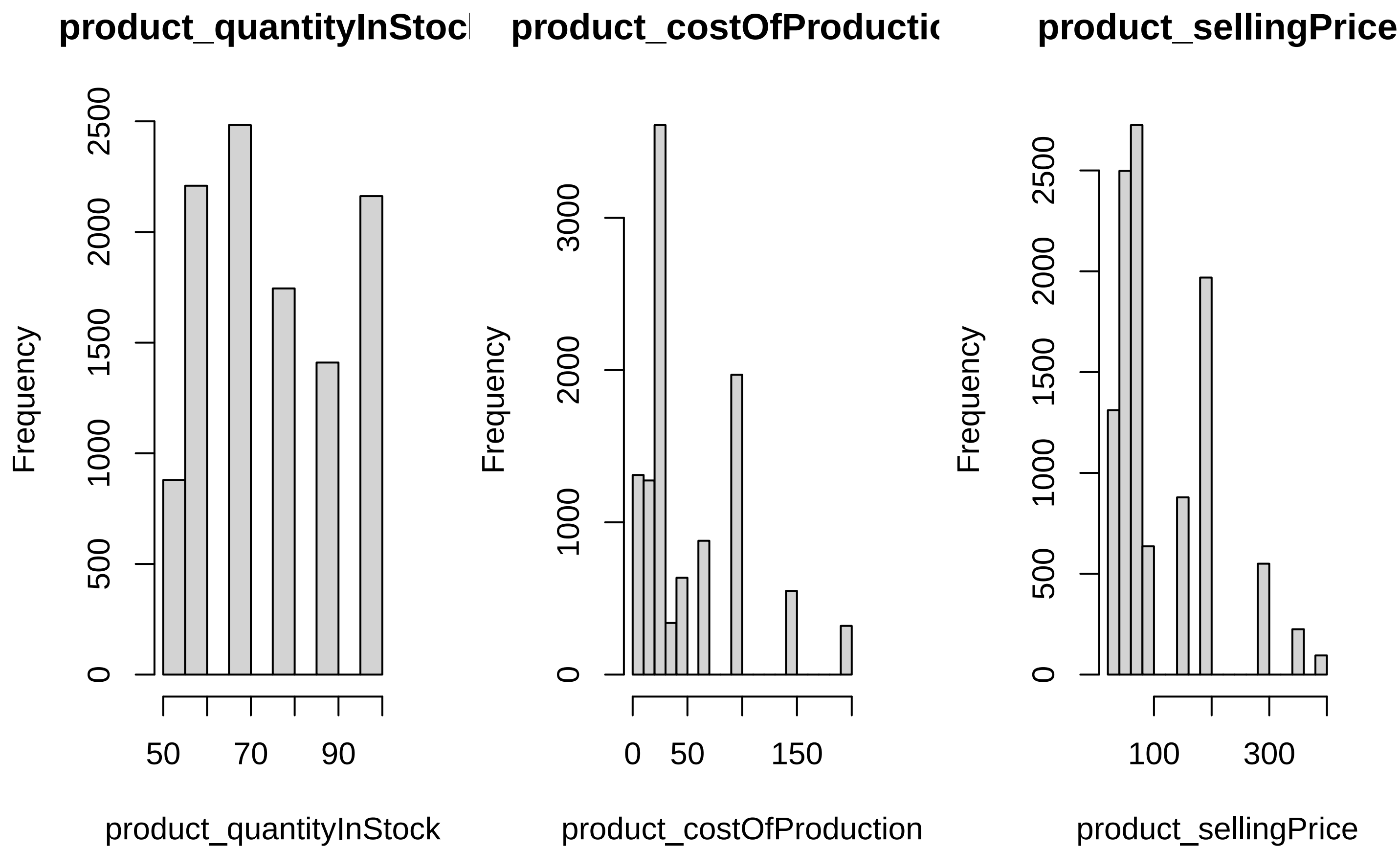
2.5.1 Histogram

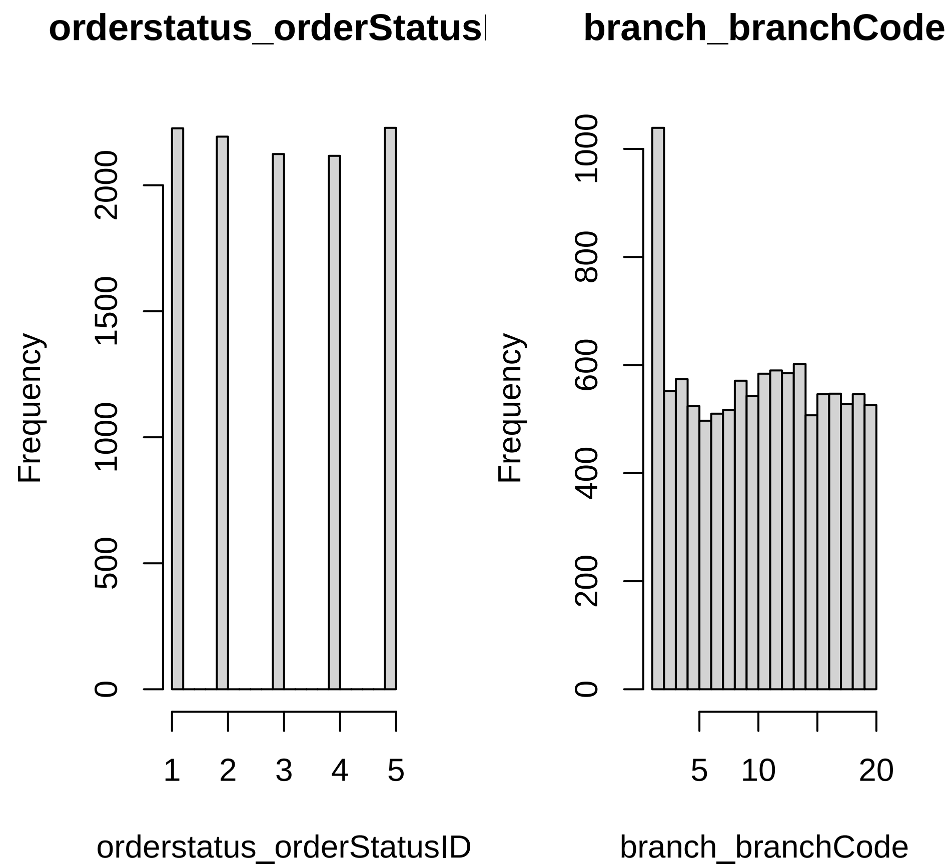
```
par(mfrow = c(1, 2))
for (i in 1:43) {
  if (is.numeric(customer_feedback_data[[i]])) {
    hist(customer_feedback_data[[i]],
         main = names(customer_feedback_data)[i],
         xlab = names(customer_feedback_data)[i])
  } else {
    message(paste("Column", names(customer_feedback_data)[i],
                  "is not numeric and will be skipped."))
  }
}
```


customer_customerNumb
customer_postalCode
customerorder_orderNum
Frequency
0
200
400
600
800
1000
0
400
800
40000
2000
5000
customer_customerNumber
customer_postalCode
customerorder_orderNumbe


customerorder_customerNu
omerfeedback_customerfee
customerfeedback_foodqu
Frequency
0
200
400
600
800
1000
1500
2000
2500
500
0
400
800
2000
5000
1
2
3
4
5
customerorder_customerNumb
customerfeedback_customerfeedb
customerfeedback_foodqualit

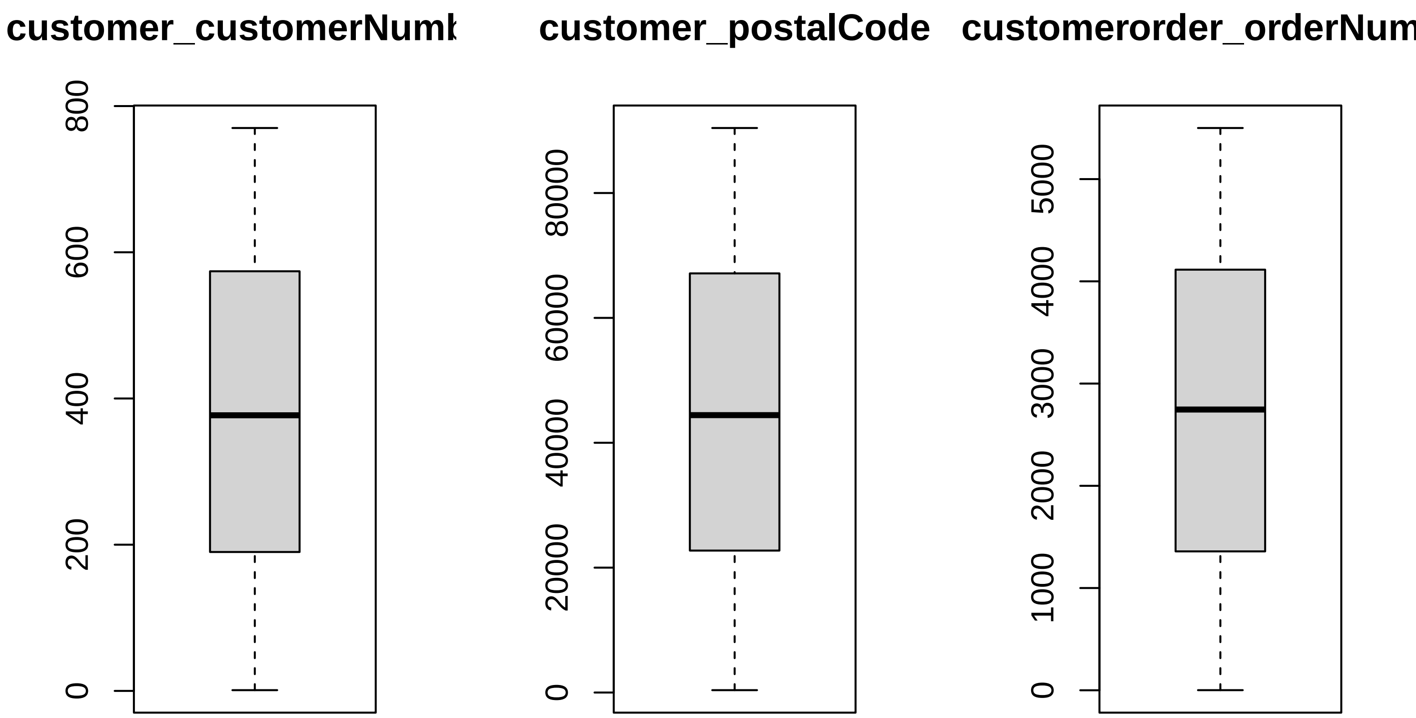

customerfeedback_pricetov
Frequency
0
500
1000
1500
2000
2500
1
2
3
4
5
customerfeedback_pricetovalu
customerfeedback_ambiar
Frequency
0
500
1000
1500
2000
2500
1
2
3
4
5
customerfeedback_ambiance
orderdetail_quantityOrder
Frequency
0
200
600
1000
1400
1
3
5
7
orderdetail_quantityOrdered

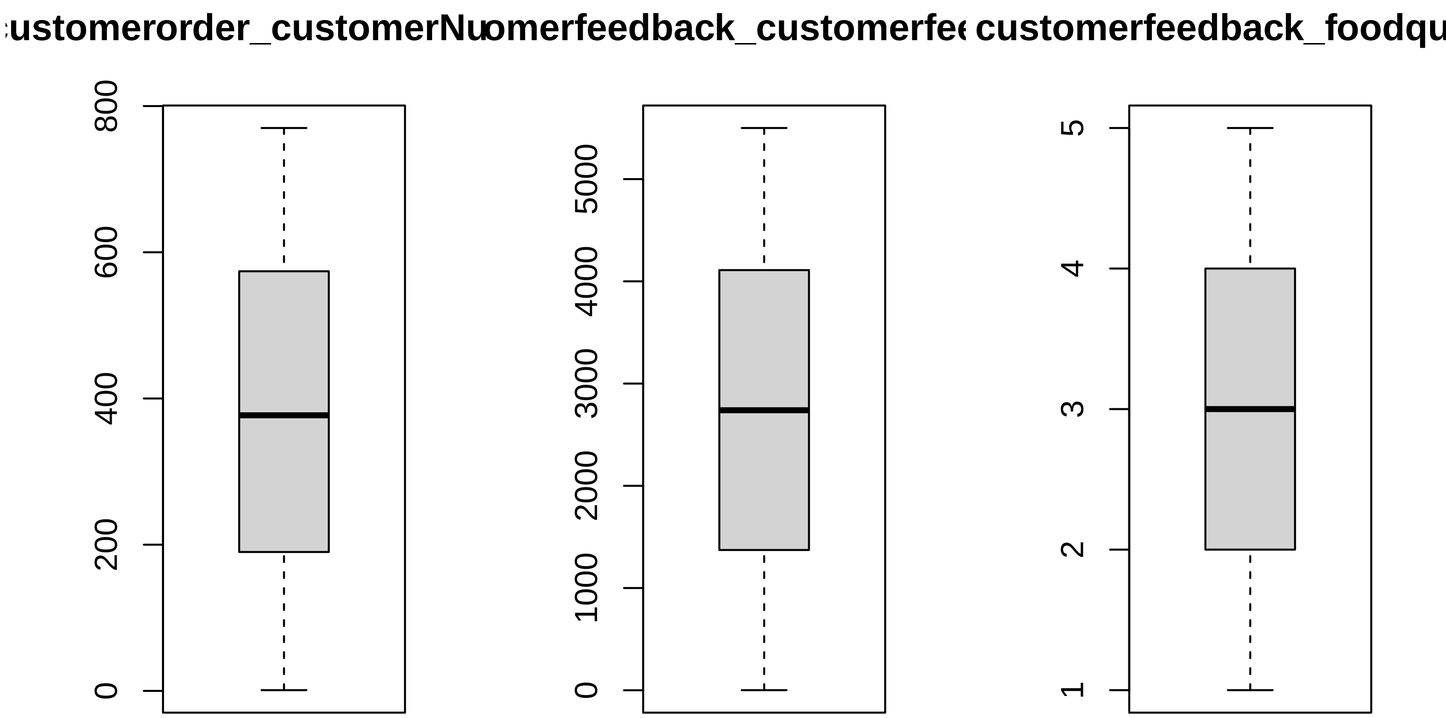

product_quantityInStock
Frequency
0
500
1000
1500
2000
2500
50
70
90
product_quantityInStock
product_costOfProductio
Frequency
0
1000
2000
3000
0
50
150
product_costOfProduction
product_sellingPrice
Frequency
0
500
1000
1500
2000
2500
100
300
product_sellingPrice



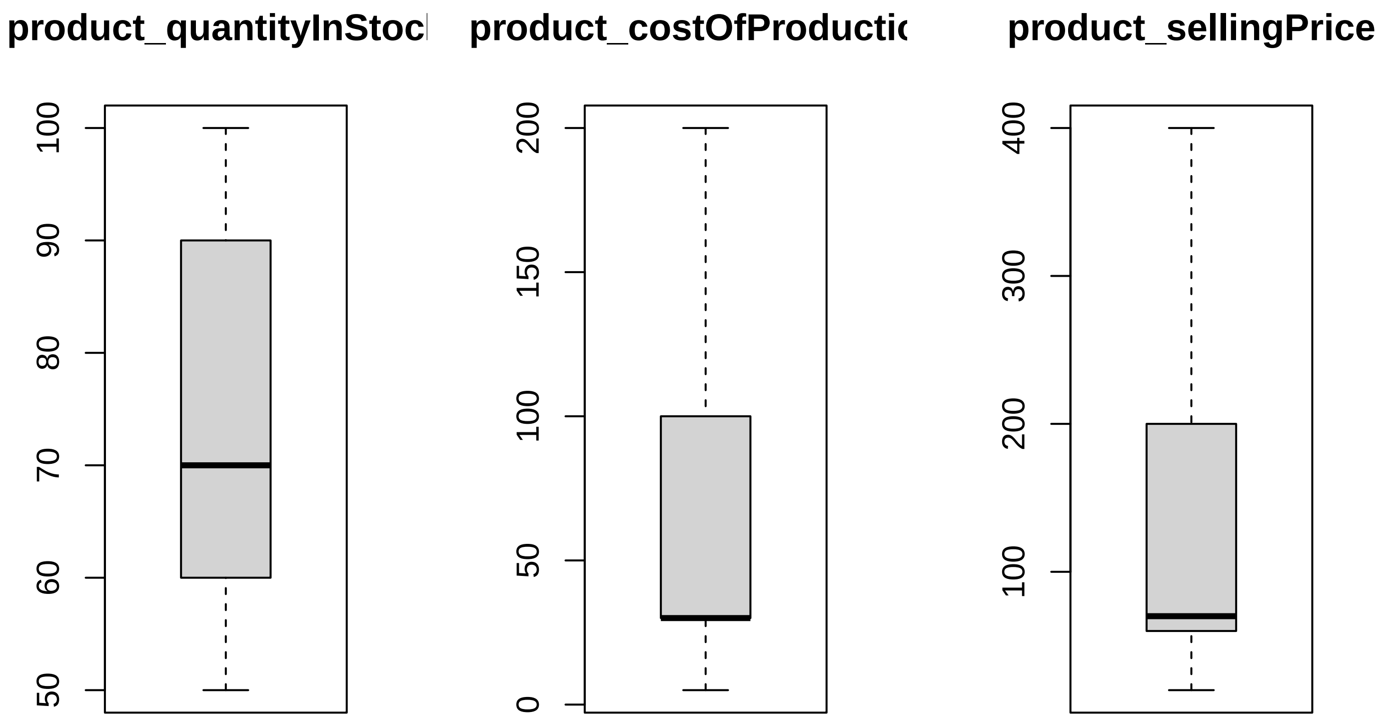
2.5.2 Box and Whisker Plot

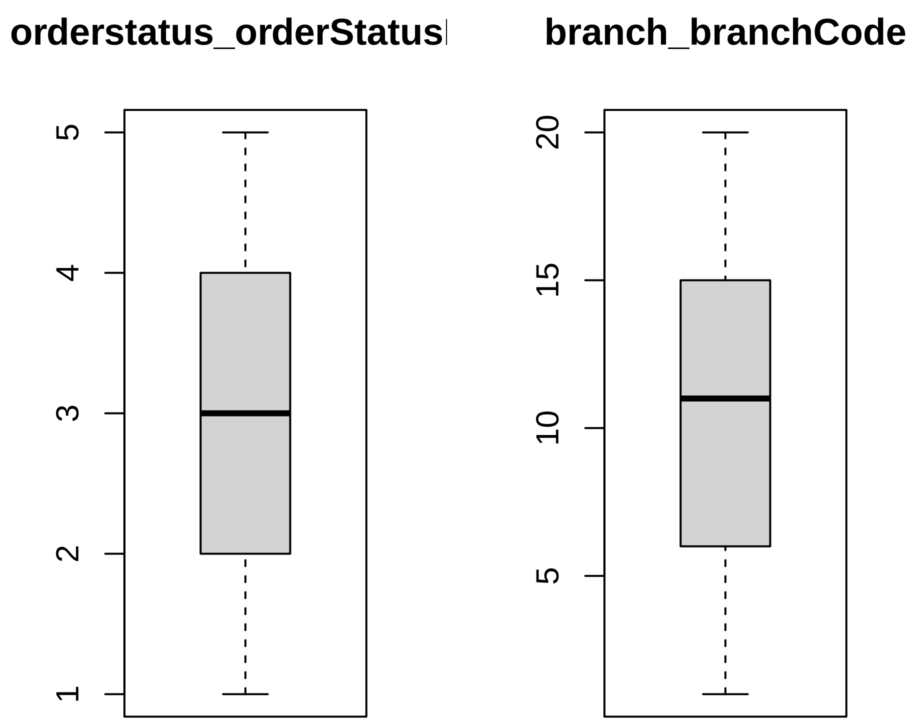
```
par(mfrow = c(1, 2))
for (i in 1:43) {
  if (is.numeric(customer_feedback_data[[i]])) {
    boxplot(customer_feedback_data[[i]], main = names(customer_feedback_data)[i])
  } else {
    message(paste("Column", names(customer_feedback_data)[i],
                  "is not numeric and will be skipped."))
  }
}
```


customer_customerNumb
800
600
400
200
0
customer_postalCode
80000
60000
40000
20000
0
customerorder_orderNum
5000
4000
3000
2000
1000
0


customerorder_customerNu
omerfeedback_customerfee
customerfeedback_foodqu
0
200
400
600
800
0
1000
2000
3000
4000
5000
1
2
3
4
5


customerfeedback_pricetov
customerfeedback_ambiar
orderdetail_quantityOrder
5
4
3
2
1
5
4
3
2
1
8
7
6
5
4
3
2
1


product_quantityInStock
100
90
80
70
60
50
product_costOfProductio
200
150
100
50
0
product_sellingPrice
400
300
200
100

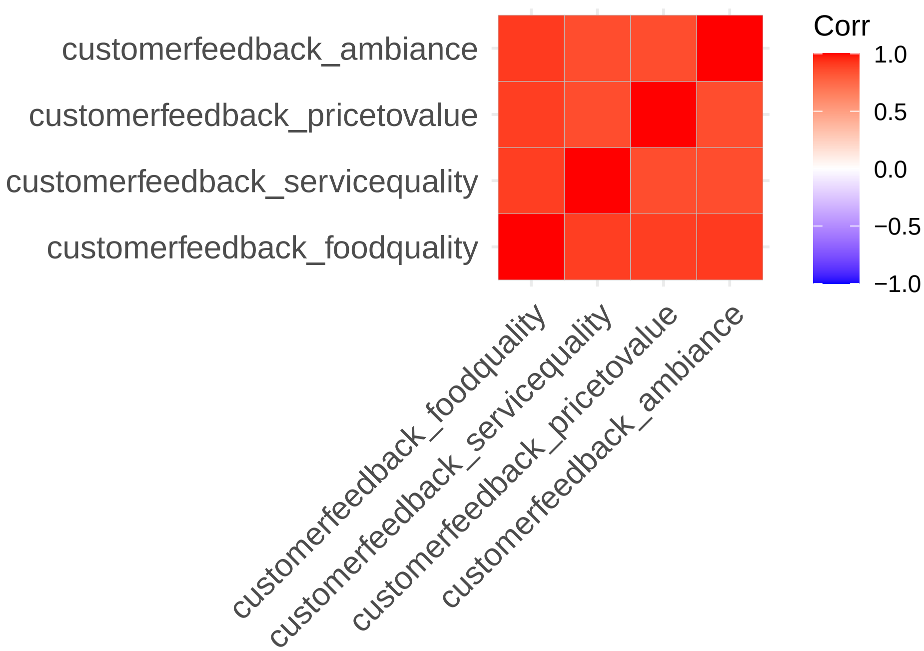


2.5.3 Missing Data Plot

```
pacman::p_load("Amelia")

missmap(customer_feedback_data, col = c("red", "grey"), legend = TRUE)
```

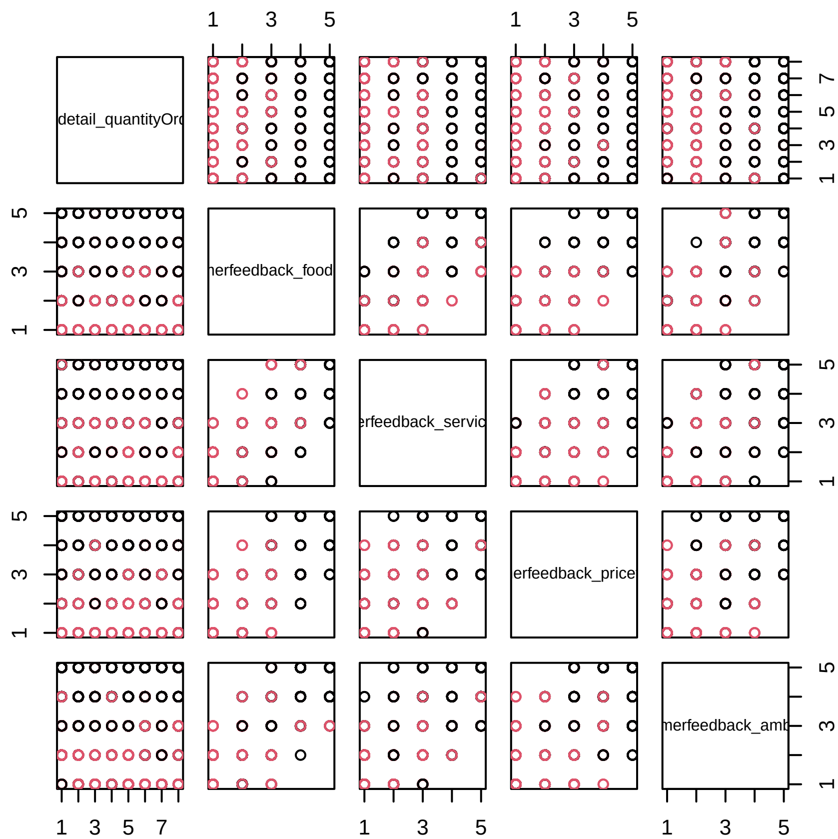
2.5.4 Correlation Plot

```
pacman::p_load("ggcorrplot")

ggcorrplot(cor(customer_feedback_data[,c(19,20,21,22)]))
```

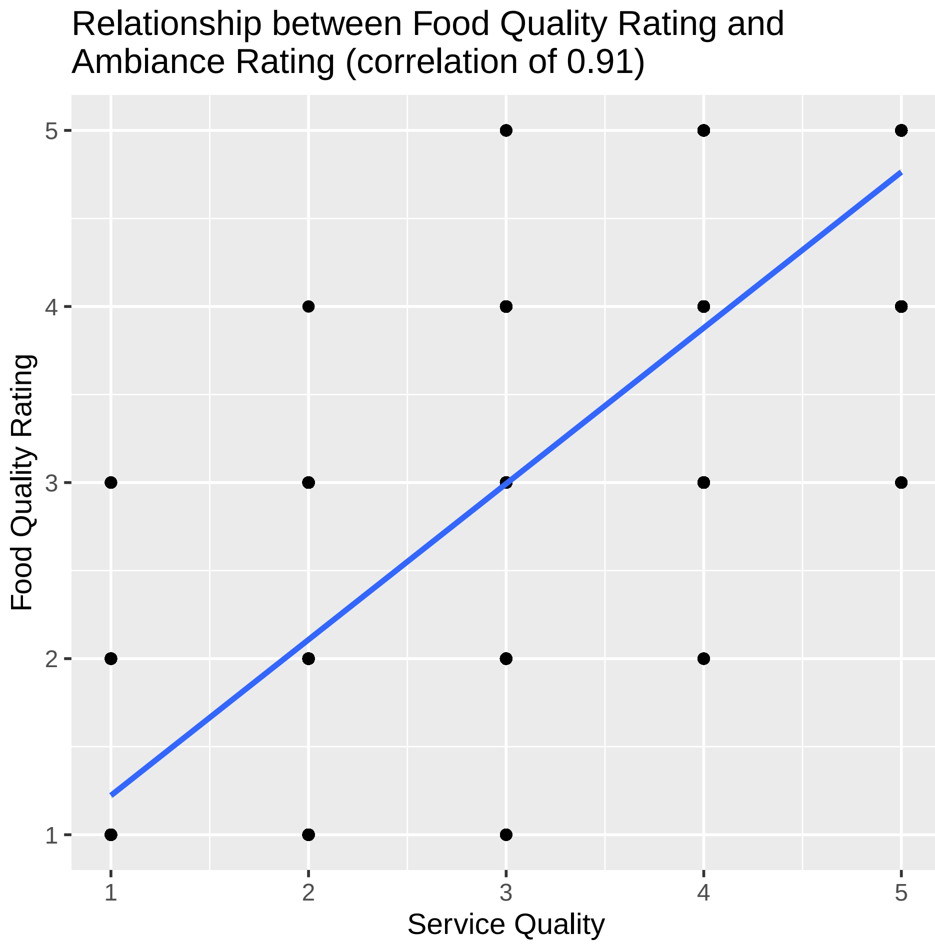


2.5.5 Scatter Plot

```
pacman::p_load("corrplot")

pairs(orderdetail_quantityOrdered ~ customerfeedback_foodquality + customerfeedback_servicequality + cus
```




```r
pacman::p_load("ggplot2")
ggplot(customer_feedback_data,
       aes(x = customerfeedback_ambiance, y = customerfeedback_foodquality)) +
  geom_point() +
  geom_smooth(method = lm) +
  labs(
    title = "Relationship between Food Quality Rating and \nAmbiance Rating (correlation of 0.91)",
    x = "Service Quality",
    y = "Food Quality Rating"
  )
```




```r
pacman::p_load("ggplot2")
ggplot(customer_feedback_data,
       aes(x = customerfeedback_ambiance, y = customerfeedback_foodquality, color = customer_status)) +
  geom_point() +
  geom_smooth(method = lm) +
  labs(
    title = "Relationship between Food Quality Rating and \n Ambiance Rating for Each Customer Status",
    x = "Service Quality Rating",
    y = "Food Quality Rating",
    color = "Customer Status"
  ) +
  scale_color_discrete(
    labels = c("1" = "Active Customer", "0" = "Dormant Customer")
  ) +
  theme_minimal()  # Optional: applies a cleaner theme
```

3 Statistical Test

We then apply a logistic regression as a statistical test for regression.

```
log_test <- glm(customer_status ~
                  customerfeedback_foodquality +
                  customerfeedback_servicequality +
                  customerfeedback_pricetovalue +
                  customerfeedback_ambiance,
                # This specifies that the generalized linear model to use
                # is logistic regression
                family = binomial,
                data = customer_feedback_data)
```

View the summary of the model.

```
summary(log_test)
```

```
##
## Call:
```

```
## glm(formula = customer_status ~ customerfeedback_foodquality +
##     customerfeedback_servicequality + customerfeedback_pricetovalue +
##     customerfeedback_ambiance, family = binomial, data = customer_feedback_data)
##
## Coefficients:
##                                 Estimate Std. Error z value Pr(>|z|)
## (Intercept)                      3.89359    0.07582  51.354  < 2e-16 ***
## customerfeedback_foodquality    -0.59450    0.05847 -10.168  < 2e-16 ***
## customerfeedback_servicequality -0.22102    0.04090  -5.404 6.50e-08 ***
## customerfeedback_pricetovalue   -0.24606    0.04103  -5.996 2.02e-09 ***
## customerfeedback_ambiance       -0.26982    0.04148  -6.506 7.74e-11 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 15094  on 10887  degrees of freedom
## Residual deviance: 10328  on 10883  degrees of freedom
## AIC: 10338
##
## Number of Fisher Scoring iterations: 4
```

The logistic regression equation is in the following form:

$$\text{log-odds(customerstatus)} = \beta_0 + \beta_1 \cdot foodquality + \beta_2 \cdot servicequality + \beta_3 \cdot pricetovalue + \beta_4 \cdot ambiance$$

Plugging in the coefficients from the output gives:

$$\text{log-odds(customerstatus)} = 3.89 - 0.59 \cdot foodquality - 0.22 \cdot servicequality - 0.25 \cdot pricetovalue - 0.27 \cdot ambiance$$

The log-odds is then converted into a probability using the logistic function as:

$$P(\text{customerstatus} = 1) = \frac{1}{1 + e^{-\text{log-odds}}}$$

- If P(customer_status=1) 0.5, predict active customer (1).
- If P(customer_status=1)<0.5, predict dormant customer (0).

Example 1: A food quality rating of 2, service quality rating of 1, a price-to-value rating of 2, and an ambiance rating of 1 results in P(customer_status=1) = 0.85 which is an active customer:

```r
coefs <- coef(log_test)
log_odds   <- coefs["(Intercept)"] +
  coefs["customerfeedback_foodquality"] * 2 +
  coefs["customerfeedback_servicequality"] * 1 +
  coefs["customerfeedback_pricetovalue"] * 2 +
  coefs["customerfeedback_ambiance"] * 1

p_manual <- 1 / (1 + exp(-log_odds))

print(p_manual)
```

```
## (Intercept)
##   0.8483414
```

Example 2: A food quality rating of 4, service quality rating of 4, a price-to-value rating of 3, and an ambiance rating of 5 results in P(customer_status=1) = 0.19 which is a dormant customer:

```r
coefs <- coef(log_test)
log_odds   <- coefs["(Intercept)"] +
  coefs["customerfeedback_foodquality"] * 4 +
  coefs["customerfeedback_servicequality"] * 4 +
  coefs["customerfeedback_pricetovalue"] * 3 +
  coefs["customerfeedback_ambiance"] * 5
p_manual <- 1 / (1 + exp(-log_odds))

print(p_manual)
```

```
## (Intercept)
##   0.1891207
```

3.1 The χ^2 Statistic and its p-Value

To obtain the p-value of the χ^2 statistic:

```r
# chi_2 <- log_test$null.deviance/1 - log_test$residuals
chi_2 <- 15094 - 10328

print(chi_2)
```

```
## [1] 4766
```

```r
# To get the degrees of freedom:
calculated_df = length(coef(log_test)) - 1  # Subtract 1 for intercept

p <- 1 - pchisq(chi_2, df = calculated_df)
# to format as a scientific notation with four digits after the decimal place
# sprintf("%.4e", p)
format.pval(p, eps = .Machine$double.eps, digits = 2)
```

```
## [1] "<2e-16"
```

3.2 95% Confidence Interval of the Parameters

To obtain a 95% confidence interval for the parameters:

```r
confint(log_test, level = 0.95)
```

```
##                                     2.5 %     97.5 %
## (Intercept)                      3.7462868  4.0435355
## customerfeedback_foodquality    -0.7093200 -0.4800961
## customerfeedback_servicequality -0.3012456 -0.1409180
## customerfeedback_pricetovalue   -0.3265687 -0.1656960
## customerfeedback_ambiance       -0.3511999 -0.1885993
```

3.3 Odds Ratio

Exponentiating the coefficients yields odds ratios, which quantify the multiplicative change in odds for a one-unit increase in the predictor.

```
exp(coef(log_test))
```

```
##                     (Intercept)    customerfeedback_foodquality
##                      49.0869338                       0.5518404
## customerfeedback_servicequality   customerfeedback_pricetovalue
##                       0.8017011                       0.7818792
##       customerfeedback_ambiance
##                       0.7635172
```

3.3.1 95% Confidence Interval for the Odds Ratio

To obtain a 95% confidence interval for the Odds Ratio:

```
pacman::p_load("dplyr")
confint(log_test) %>% exp()
```

```
##                                      2.5 %     97.5 %
## (Intercept)                     42.3634840 57.0276055
## customerfeedback_foodquality     0.4919786  0.6187240
## customerfeedback_servicequality  0.7398960  0.8685605
## customerfeedback_pricetovalue    0.7213948  0.8473038
## customerfeedback_ambiance        0.7038430  0.8281182
```

3.4 Akaike Information Criterion (AIC)

AIC specifies how well a model fits the data from which it was generated. The AIC value is calculated using the number of predictor variables and also the estimate of the maximum likelihood of the model. AIC penalizes complexity, therefore, any model with a lesser AIC value is a more significant model. A lower AIC value means the complexity of the model is lower and the model better explains the variations. The model's AIC value of `10,338` can be compared with the AIC value of alternative models, e.g., a logistic regression model that has dropped one of the original predictors.

3.5 McFadden's Pseudo R^2

Logistic regression does not use the traditional R^2, but we can compute a **pseudo-R^2** (McFadden's) as follows:

```
pseudo_r2 <- 1 - (log_test$deviance / log_test$null.deviance)
print(pseudo_r2)
```

```
## [1] 0.3157299
```

We get a McFadden's pseudo-R^2 of `0.32` . A pseudo-R^2 > 0.2 is considered strong in logistic regression.

3.6 Fisher Scoring Iterations

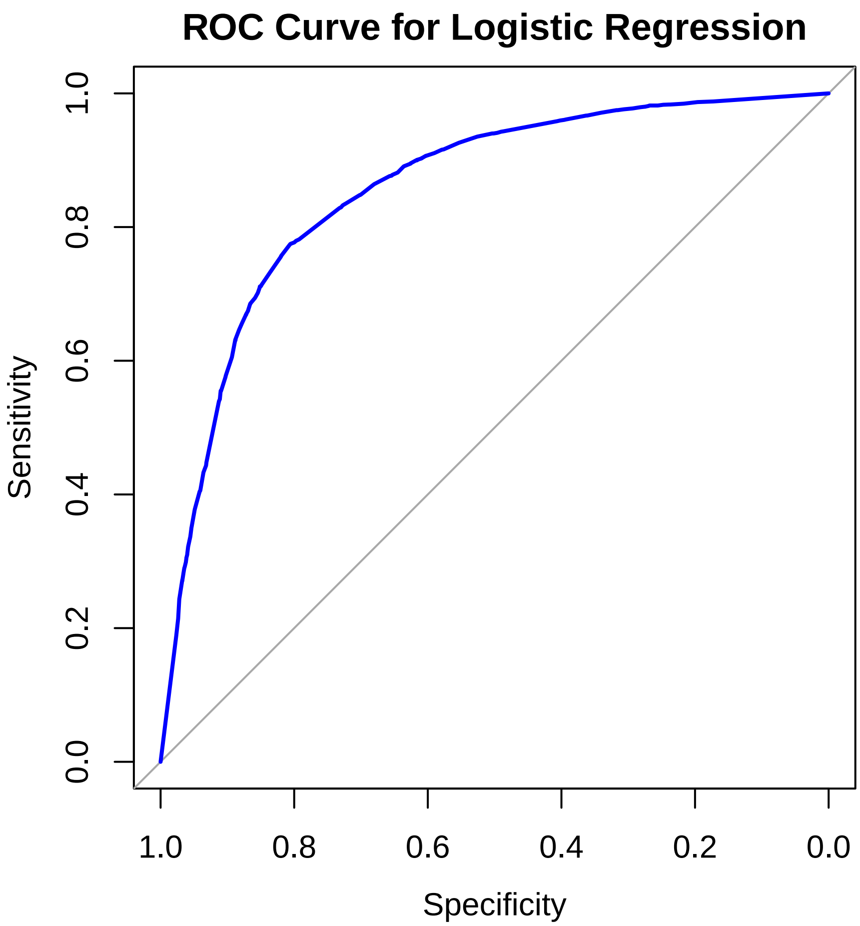
There were 4 Fisher scoring iterations which indicates that the fitting routine required 4 iterative updates to reach convergence. A small number (< 10) suggests that the algorithm found the maximum-likelihood estimates efficiently without convergence warnings.

3.7 Model Fit Metrics

The ROC Curve and the AUC gives insight into how well the logistic regression model distinguishes between the two outcome classes (1 = active customer and 0 = dormant customer).

- The x-axis shows the False Positive Rate (FPR), or specificity.
- The y-axis shows the True Positive Rate (TPR), or sensitivity.
- The curve shows how TPR and FPR change as the decision threshold changes from 0 to 1.
- A curve closer to the top-left corner indicates better model performance.

```
pacman::p_load("pROC")

predicted_probs <- predict(log_test, type = "response")

roc_obj <- roc(customer_feedback_data$customer_status, predicted_probs)
auc_value <- auc(roc_obj)
plot(roc_obj, col = "blue", main = "ROC Curve for Logistic Regression")
```




```r
print(auc_value)
```

```
## Area under the curve: 0.855
```

Table 2: AUC Value Interpretation

AUC Value	Interpretation
0.5	Using the logistic regression model is equivalent to randomly guessing
0.6 - 0.7	Poor discrimination of the classes
0.7 - 0.8	Acceptable discrimination of the classes (fair)
0.8 - 0.9	Good discrimination
0.9 - 1.0	Excellent discrimination

An AUC of `0.86` indicates that the model has a good ability to discriminate between the two classes.

4 Diagnostic EDA

Diagnostic EDA is performed to validate that the regression assumptions are true with respect to the statistical test. Validating the regression assumption in turn ensures that the statistical tests applied are appropriate for the data and helps to prevent incorrect conclusions.

4.1 Test of Linearity

The test of linearity is necessary given that linearity is one of the key assumptions of statistical tests of regression and verifying it is crucial for ensuring the validity of the model's estimates and predictions.

Component-Plus-Residual (Partial-Residual) Plots

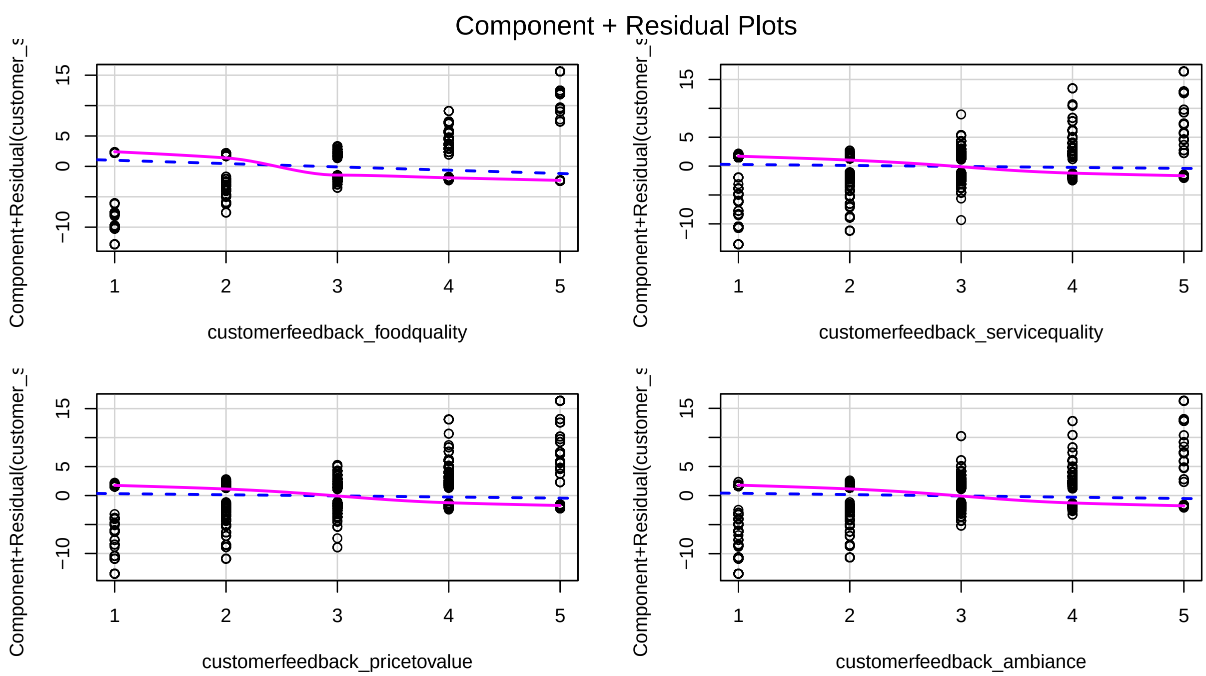
Logistic regression assumes that the relationship between the log-odds of the outcome and the continuous predictors is linear. A log-odds is the natural logarithm of the odds of an event occurring.

If p is the probability of an event (where $0 < p < 1$), the odds of the event occuring is the ratio of the probability that the event occurs to the probability that the event does not occur, i.e., $\frac{p}{1-p}$ and the log-odds (computed using the logit function) of the event, p, is:

$$\text{logit}(p) = \ln\left(\frac{p}{1-p}\right)$$

A roughly straight line indicates the "log-odds - predictor" linearity assumption is met.

```
pacman::p_load("car")

crPlots(log_test)
```



4.2 Test of Independence of Errors (Autocorrelation)

This test is necessary to confirm that each observation is independent of the other. It helps to identify **autocorrelation** that is introduced when the data is collected over a close period of time or when one observation is related to another observation. Autocorrelation leads to underestimated standard errors and inflated test statistics. It can also make findings appear more significant than they actually are.

The "**Durbin-Watson Test**" can be used as a test of independence of errors (test of autocorrelation). A Durbin-Watson statistic close to 2 suggests no autocorrelation, while values approaching 0 or 4 indicate positive or negative autocorrelation, respectively.

- The null hypothesis, H_0, is that there is no autocorrelation
- The alternative hypothesis, H_a, is that there is autocorrelation

If the p-value of the Durbin-Watson statistic is greater than 0.05 then there is no evidence to reject the null hypothesis that "there is no autocorrelation". The results below show a p-value of $< .001$, therefore, the test of independence of errors around the regression line **fails**.

```
pacman::p_load("lmtest")
dwtest(log_test)
```

```
##
##  Durbin-Watson test
##
## data:  log_test
## DW = 1.0144, p-value < 2.2e-16
## alternative hypothesis: true autocorrelation is greater than 0
```

4.3 Test of Normality

Logistic regression does not assume normality of residuals or predictors.

4.4 Test of Homoscedasticity

The test of homoscedasticity is not relevant for logistic regression.

4.5 Test of Multicollinearity

Multicollinearity arises when two or more independent variables (predictors) are highly intercorrelated. The **Variance Inflation Factor (VIF)** quantifies how much the variance of a coefficient estimate is "inflated" due to multicollinearity. A VIF of 1 indicates no collinearity; values above 5 suggest problematic levels of collinearity. High VIF values (VIF $>$ 5) suggest that the coefficient estimates are less reliable due to the correlations between predictors.

```
pacman::p_load("car")
car::vif(log_test)
```

```
##    customerfeedback_foodquality customerfeedback_servicequality
##                        6.055689                        3.449586
##   customerfeedback_pricetovalue       customerfeedback_ambiance
##                        3.355231                        3.477957
```

4.6 Test of Outliers

The `influencePlot()` function in R combines 3 key diagnostic measures into a single plot to identify influential observations.

The plot displays:

- Y-axis: Studentized residuals (standardized residuals adjusted for leverage). This measures "outlierness" in the outcome variable.
- X-axis: Leverage (hat values), measuring how "unusual" an observation is in terms of its predictor values.
- Bubble size: Cook's distance, quantifying the influence of each observation on the model coefficients.

Top-left/bottom-left:

- Indicates observations with high residuals but low leverage: Outliers in the outcome but not predictors.
- Possible next step: Investigate the observations for misclassified outcomes.

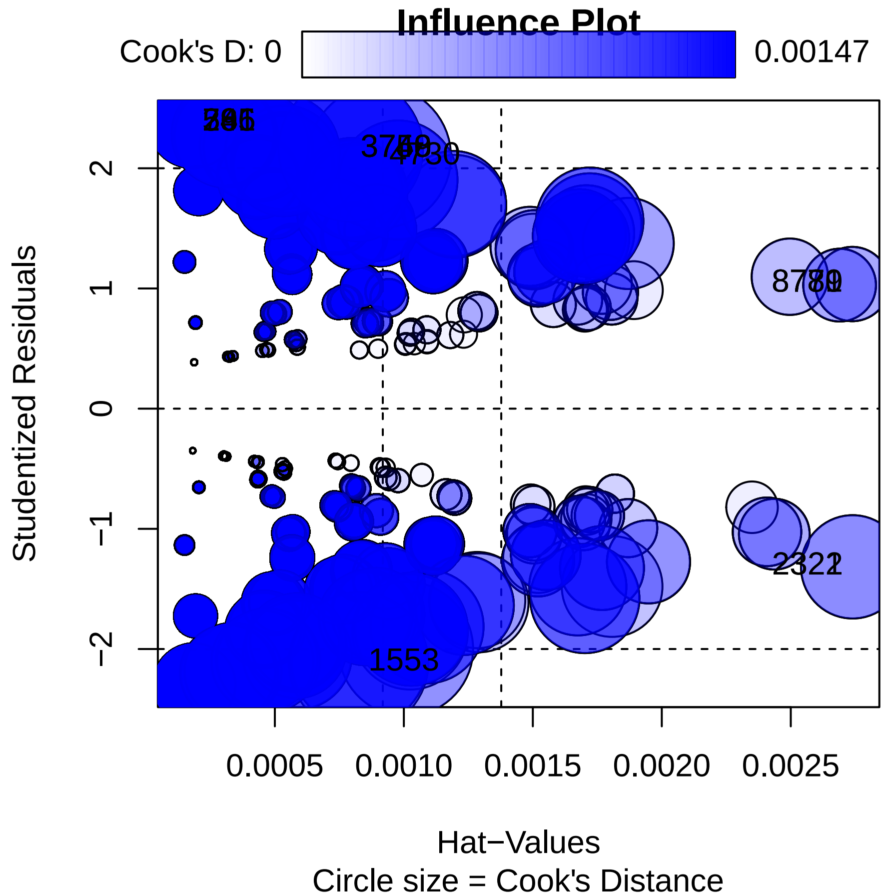
Top-right/bottom-right:

- Indicates high residuals and high leverage: Influential outliers that distort the model.
- Possible next step: These are the most problematic observations. You need to check if they are valid data points or errors.

Middle-right:

- High leverage but residuals near 0: Unusual predictor values but well-predicted outcomes.
- Possible next step: These observations are generally safe to keep.

```r
pacman::p_load("car")

influencePlot(log_test,
              id = list(n = 5),  # Label top 5 influential points
              main = "Influence Plot",
              sub = "Circle size = Cook's Distance")
```

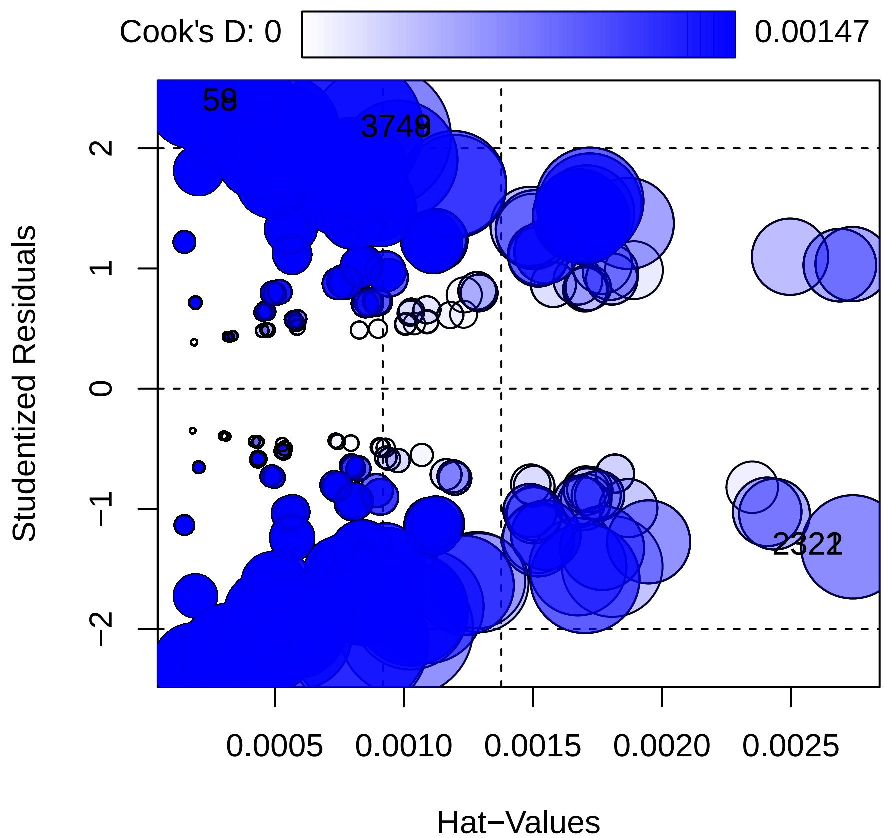



```
##        StudRes          Hat        CookD
## 58    2.377381 0.0001816594 0.0005761792
## 59    2.377381 0.0001816594 0.0005761792
## 265   2.377381 0.0001816594 0.0005761792
## 266   2.377381 0.0001816594 0.0005761792
## 741   2.377381 0.0001816594 0.0005761792
## 1553 -2.115156 0.0008249546 0.0013770576
## 2321 -1.316394 0.0027398818 0.0007570392
## 2322 -1.316394 0.0027398818 0.0007570392
## 3748  2.158768 0.0007948046 0.0014713998
## 3749  2.158768 0.0007948046 0.0014713998
## 3750  2.158768 0.0007948046 0.0014713998
## 4730  2.096247 0.0009061460 0.0014464124
## 8779  1.036695 0.0027386845 0.0003909379
## 8780  1.036695 0.0027386845 0.0003909379
## 8781  1.036695 0.0027386845 0.0003909379
```

The influential outliers can then be:

1. Corrected for data entry errors, e.g., an age of 240 years
2. Deleted so that the statistical test can be run again without them

```r
influential_points <- as.numeric(row.names(influencePlot(log_test)))
```




```r
customer_feedback_data_infl <- customer_feedback_data[influential_points, ]
```

Print the observation numbers that have been identified as influential outliers

```r
print(influential_points)
```

```
## [1]   58   59 2321 2322 3748 3749
```

Print the influential observations' predictor and outcome values.

```r
head(customer_feedback_data_infl)
```

```
## # A tibble: 6 x 43
##   customer_customerNumber customer_customerName    customer_contactFirstName
##                     <dbl> <chr>                    <chr>
## 1                     498 [Business] Serena Inn    Emmanuel
## 2                     498 [Business] Serena Inn    Emmanuel
## 3                     321 [Business] Diani Hotel   Pierre
```

```
## 4                     321 [Business] Diani Hotel   Pierre
## 5                     263 [Business] Voyager Haven Kiptoo
## 6                     263 [Business] Voyager Haven Kiptoo
## # i 40 more variables: customer_contactLastName <chr>, customer_phone <chr>,
## #   customer_addressLine1 <chr>, customer_addressLine2 <chr>,
## #   customer_postalCode <dbl>, customer_county <chr>, customer_subCounty <chr>,
## #   customer_status <fct>, customerorder_orderNumber <dbl>,
## #   customerorder_orderDate <dttm>, customerorder_requiredDate <dttm>,
## #   customerorder_dispatchDate <dttm>, customerorder_orderStatusID <dbl>,
## #   customerorder_customerNumber <dbl>, ...
```

5 Interpretation of the Results

5.1 Academic Statement

A logistic regression analysis was conducted on the customer feedback data (N = 10,888) to examine whether food quality rating, service quality rating, price to value rating, and ambiance rating predicted a customer's status as active or dormant. The customer's status was coded as 1 for active and 0 for dormant.

Subtracting the residual deviance (15,094 on 10,887 df) from the null deviance (10,328 on 10,883 df) gave a $\chi^2(4, \text{N} = 10{,}888) = 4{,}766$, $p < .001$, thus showing that the model was statistically significant compared to the null model. An Area Under the Curve (AUC) value of `0.86` indicated that the model has a good ability to discriminate between the two classes.

The results are reported in the table below:

Table 3: Regression Coefficients Predicting a customer's status from customer feedback and other predictors

Predictor	β	95% CI	SE	z	p
(Intercept)	3.89	[3.75, 4.04]	0.08	51.354	$< .001$
Food Quality	-0.59	[-0.71, -0.48]	0.06	-10.17	$< .001$
Service Quality	-0.22	[-0.30, -0.14]	0.04	-5.40	$< .001$
Price to Value	-0.25	[-0.33, -0.17]	0.04	-6.00	$< .001$
Ambiance	-0.27	[-0.35, -0.19]	0.04	-6.51	$< .001$

Note. N = 11,026; SE = standard error; CI = confidence interval.

Significant Predictor Effects

Assumption: The feedback scales are such that 1 = Excellent and 5 = Poor

- The customer feedback rating for the quality of food (β = -0.59, 95% CI [-0.71, -0.48], SE = 0.06, z = -10.17, $p < .001$) was identified as the most significant predictor. A one-unit increase in the "customerfeedback_pricetovalue" predictor is associated with a 45% decrease in the odds of a customer being active ($OR = 0.55$).
- The customer feedback rating for the ambiance (β = -0.27, 95% CI [-0.35, -0.19], SE = 0.04, z = -6.51, $p < .001$) was identified as the second-most significant predictor. A one-unit increase in the "customerfeedback_ambiance" predictor is associated with a 24% decrease in the odds of a customer being active ($OR = 0.76$).

- While the customer feedback rating for the service quality and the price to value are both significant, they exhibit relatively smaller effects on the outcome. A one-unit increase in each is associated with a 20% decrease and 22% decrease in the odds of a customer being active respectively ($OR = 0.80$ and $OR = 0.78$ respectively).

5.2 Business Analysis

Key insights:

1. Marketing and loyalty campaigns may be optimised by focusing outreach on at-risk branches identified by poor food-quality and ambiance ratings.
2. Investment in menu refinement and consistency should be considered to improve food quality ratings and thereby enhance customer retention.
3. Enhancements to the dining environment—such as decor updates, lighting adjustments, and noise control—should be considered to improve ambiance perceptions.

5.3 Limitations

1. Assumes no clustering (e.g., multiple feedback per customer) or autocorrelation (e.g., time-series trends).
2. The Durbin-Watson test fails therefore there is autocorrelation in the data, i.e., there is a correlation between residuals across observations.
3. High Variance Inflation Factor (VIF) values (VIF $>$ 5) suggest that the coefficient estimates are less reliable due to the correlations between predictors. `customerfeedback_foodquality` has a VIF = 6.01